Do LLM Cascades Need Smart Stopping? A Real-Data Audit of Deferral Rules

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Abstract

An LLM cascade sends a query to a cheap model first and escalates to larger models only when needed; its *deferral rule* decides when to stop. Increasingly sophisticated rules have been proposed, often evaluated in simulation or against a single global confidence threshold. We audit seven rules on real per-question log-likelihoods of 13 open models from three families (Qwen1.5 0.5B–72B, Llama-2 7B–70B, Mistral/Mixtral) on 16,481 questions from MMLU, ARC-Challenge and WinoGrande, taken from the public Open LLM Leaderboard logs. Under a protocol in which every rule is tuned on held-out data only, we find: (i) moving from one global threshold to per-stage thresholds gains 0.8–4.3 AUCC points, more than any further, more sophisticated rule adds on top; (ii) choosing *which* models to chain (a subset cascade) is best in two of three families, degrades least with few labels, and survives task shift, reaching 99% of the largest model's accuracy at 74% (Qwen) and 81% (Mistral) of its cost; (iii) a value-of-computation lookahead rule that looked strong in our own earlier simulation is never the best rule on real data, and its category conditioning hurts in two of three families; (iv) in a controlled sweep, adaptive rules overtake simple model selection only once confidence AUROC exceeds roughly 0.9, above today's real signals (0.74–0.81), at which point matching the largest model can cost as little as 8% of it. The practical lesson is to tune thresholds per stage, choose the model subset, and invest in confidence quality before stopping-rule sophistication. Code, data-fetching scripts and all results are released.

1 Introduction

Serving every query with the largest model is expensive, yet many queries are answered correctly by much smaller ones. *Cascades* exploit this: a query goes to the cheapest model first and is escalated to larger models only when a deferral rule judges the current answer unreliable [1, 2]. The rule is the crux of the design, and the literature has moved from confidence thresholds [1] to learned post-hoc deferral [2, 3], answer-consistency signals [4], provably optimal cascade routing [5], decision-theoretic characterisations [6] and online optimal-stopping formulations [7]; a recent survey maps the space [8].

Two evaluation habits make it hard to know how much this sophistication buys in practice. First, new rules are often compared against a *single global threshold* shared by all stages, a baseline that ignores that escalation costs differ sharply between stages. Second, evidence is often simulated or drawn from few models. We ran into both ourselves: in unpublished simulation experiments, a value-of-computation lookahead rule we proposed (VoC-Stop, §4) appeared to cut cost by 17–74% at matched accuracy against a global threshold. When we added a per-stage baseline inside the same simulator, most of that saving disappeared, and on real data the rule was never the best.

This paper is the audit we should have run first. We use public per-question log-likelihoods for 13 open models [12], which let us evaluate any cascade offline, for free, on real correctness and real confidence. Our contributions are:

1. A reproducible real-data benchmark for deferral rules: three model families, 16,481 questions, one protocol under which every rule is tuned on held-out data only (§3).
2. Five experiments: main comparison, confidence quality, label budget, task shift and an ablation (§5), with four findings summarised in the abstract.
3. A measured account of why a simulator overstated a method's value (§6), and practical guidance for practitioners (§7).

2 Related Work

Cascades and deferral. FrugalGPT [1] popularised LLM cascades with learned score thresholds. Jitkrittum et al. [2] characterise the optimal deferral rule and show when confidence-based deferral suffices and when post-hoc learned deferral helps; Gupta et al. [3] extend this to generative LMs using token-level uncertainty. Yue et al. [4] use answer consistency of the weaker model as the deferral signal. Dekoninck et al. [5] unify routing and cascading and derive an optimal strategy, identifying the quality estimator as the key factor, which our confidence sweep corroborates from a different angle. Bouchard [6] characterises cascades with constrained optimisation over eight models and concludes that performance is limited primarily by structural cost; our subset-cascade and single-model results are consistent with this. Belloni et al. [7] cast contextual cascading as an online Pandora's-box problem. RouterBench [9] standardised routing evaluation.

Our angle. We do not propose a new rule. We ask how much existing rule families gain over *properly tuned* simple baselines

on real logs, and when sophistication starts to pay. The value-of-computation view we test comes from rational metareasoning [10]; calibration uses isotonic regression [11].

3 Setup

Data. The Open LLM Leaderboard (v1) [12] evaluated models with the LM Evaluation Harness [13] and published per-question details, including the log-likelihood of every answer option. We use MMLU (5-shot) [14], ARC-Challenge (25-shot) [15] and WinoGrande (5-shot) [16] for three cascades of base models: Qwen1.5 {0.5, 1.8, 4, 7, 14, 32, 72}B [17], Llama-2 {7, 13, 70}B [18] and Mistral-7B $\rightarrow$ Mixtral-8x7B $\rightarrow$ Mixtral-8x22B [19, 20]. Questions are aligned across models by a hash of the question text, giving 16,481 questions answered by every model. Accuracies match the published leaderboard values (e.g. Llama-2-70B MMLU 69%, Mixtral-8x7B 70%).

Confidence and categories. A model's confidence is the maximum softmax probability over the option log-likelihoods. Its AUROC for predicting the model's own correctness ranges from 0.62 (Qwen1.5-0.5B) to 0.83 (Mixtral-8x22B). Categories are the four MMLU super-categories plus ARC and WinoGrande.

Cost. Per-query cost is proportional to active parameters (prompts are identical across models); Mixtral uses active parameters (12.9B, 39B). A cascade pays for every model it runs.

Protocol. Each run splits questions 50/50 into train and test. Per-stage isotonic calibration and every rule's parameters are fitted on train only; all numbers are on test, averaged over 5 random splits (3 for E2 and E3). Rules trade accuracy for cost through an exchange rate λ: at each of 30 log-spaced λ values a rule is tuned to maximise accuracy $-$ cost$/\lambda$ on train, tracing an accuracy–cost frontier on test.

Metrics. **AUCC** is the area under the (monotone envelope of the) test accuracy–cost frontier over the cost range from the cheapest model to running every model, normalised to $[0, 1]$ on the cost axis. **Cost@99** is the cost needed to reach 99% of the largest model's accuracy, as a percentage of calling the largest model alone.

4 Rules Compared

Stages $s = 1, \ldots, M$ are ordered by cost; $\hat{p}_s$ is the calibrated confidence at stage s and z the category.

- **Global threshold**: stop at the first stage with $\hat{p}_s \geq \tau$, one τ for all stages.
- **Per-stage threshold**: one τ_s per stage (as in FrugalGPT), tuned by coordinate ascent initialised at the best global threshold, so it is never worse than the global rule on train.
- **Per-stage + category threshold**: separate per-stage thresholds for each category.
- **VoC-Stop** (our earlier proposal): continue iff $\max_{t>s} [\hat{g}_{s\to t}(\hat{p}_s, z) - (C_t - C_s)/\lambda] > \hat{p}_s$, where $\hat{g}_{s\to t}$ is an isotonic estimate of stage-t accuracy given the current confidence within category z and C_s is cumulative cost: a value-of-computation lookahead rule.
- **Learned deferral (GBM)**: gradient-boosted estimates of $P(\text{stage } t \text{ correct} \mid \hat{p}_1, \ldots, \hat{p}_s, z)$ plugged into the same lookahead decision, a post-hoc deferral rule in the spirit of [2, 3].
- **Subset cascade**: for each λ, choose on train which models to run (any cost-ordered subset of at most three) and their per-stage thresholds; only models in the subset are paid for. This is a simple form of the model selection that cascade routing [5] performs; we do not reimplement their full method.
- **Single models**: call one model only (no cascade). **Oracle**: stop, per question, at the stage that is best in hindsight (upper bound).

5 Results

E1: main comparison (Table 1, Fig. 1). Three patterns hold across families. *The global threshold is the weakest cascade rule everywhere*; per-stage thresholds improve AUCC by 0.043 (Qwen), 0.019 (Llama-2) and 0.008 (Mistral), more than the best further gain any other rule achieves over per-stage thresholds (0.013, 0.016 and 0.006 respectively). *Model selection matters*: the subset cascade is best on Qwen and Mistral, typically chaining two large models (e.g. 32B$\rightarrow$72B), and reaches 99% of the largest model's accuracy at 74% and 81% of its cost. On Llama-2, where the 70B model is far more accurate than the 13B (70% vs 57%), no rule saves cost at the 99% target, and learned deferral has the best AUCC. *VoC-Stop is never the best rule*: it trails per-stage thresholds on Qwen and Mistral and trails learned deferral on Llama-2. The oracle's large margin (4–30% Cost@99) shows ample headroom, which the next experiment attributes to confidence quality.

E2: confidence quality decides whether sophistication pays (Fig. 2, Table 2). We perturb the real confidence in logit space: adding Gaussian noise degrades it; adding $\beta(2y - 1)$, where y is correctness, sharpens it. Sharpening uses the label and is not deployable; it answers "what would a better confidence estimator buy?". On Qwen and Mistral the subset cascade wins at and below real signal quality (AUROC 0.74 and 0.81), and adaptive rules (per-stage + category, learned deferral) overtake it only around AUROC 0.90–0.92. With better confidence the cost of matching the largest model collapses: on Qwen, Cost@99 falls from 73% with real confidence to 8% at AUROC 0.96. On Llama-2 the adaptive rules lead at every quality level. Three alternative real signals from the same logs (top-two margin, normalised entropy, agreement with the pre-

Table 1: E1: main comparison on real data (16,481 questions; mean ± s.d. over 5 random splits; every method tuned on the training half only). AUCC: area under the accuracy–cost curve (higher is better). Cost@99: cost to reach 99% of the largest model's accuracy, as % of calling that model alone (– = not reached in every split). Best non-oracle AUCC in bold.

	Qwen1.5 (7 models)		Llama-2 (3)		Mistral (3)	
Method	AUCC	Cost@99	AUCC	Cost@99	AUCC	Cost@99
Global threshold	0.668 ±0.008	112%	0.584 ±0.009	111%	0.726 ±0.004	98%
Per-stage threshold	0.711 ±0.002	96%	0.603 ±0.006	109%	0.734 ±0.002	94%
Per-stage + category thr.	0.713 ±0.002	101%	0.616 ±0.003	107%	0.737 ±0.002	93%
VoC-Stop	0.696 ±0.002	–	0.615 ±0.003	–	0.731 ±0.002	122%
Learned deferral (GBM)	0.708 ±0.002	–	**0.619 ±0.002**	–	0.734 ±0.003	–
Subset cascade (≤3)	**0.724 ±0.002**	74%	0.608 ±0.011	100%	**0.740 ±0.003**	81%
Single models (no cascade)	0.721 ±0.002	100%	0.593 ±0.003	100%	0.725 ±0.003	100%
Oracle (upper bound)	0.899 ±0.002	4%	0.778 ±0.003	30%	0.843 ±0.003	23%

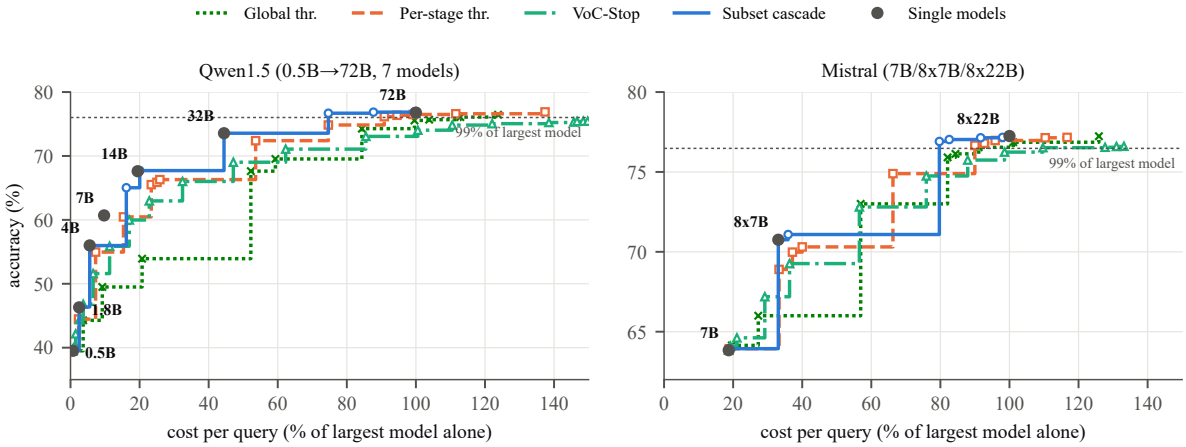


Figure 1: Accuracy–cost frontiers on real data (one test split). Dots are single models. The subset cascade (blue) dominates the full-ladder rules on Qwen1.5 and Mistral; the global threshold (green, dotted) is weakest.

vious model) all had AUROC within 0.04 of max-probability and did not change the ranking.

E3: label budget (Fig. 3). Deployments often have few labelled examples. With 100 labels the subset cascade stays within 0.03 AUCC of its full-data value in every family, whereas VoC-Stop loses 0.16 (Qwen), 0.06 (Llama-2) and 0.03 (Mistral) and category thresholds lose up to 0.13. Rules that condition on more context need more data, and on Qwen and Mistral they do not recover their deficit even with all 8,240 training questions.

E4: task shift (Table 3). Tuning on MMLU and testing on ARC and WinoGrande, the subset cascade remains best on Qwen (Cost@99 62%) and Mistral (64%). On Llama-2 all rules are within 0.03 AUCC; VoC-Stop is marginally best and the subset cascade falls back to the single largest model.

E5: which ingredients matter (Table 4). Calibrating the confidence changes little (at most +0.006 AUCC). Per-stage thresholds are the largest single step in every family. For VoC-Stop, removing category conditioning *improves* AUCC on Qwen (0.696→0.713) and Mistral (0.731→0.736) and costs 0.003 on Llama-2, contradicting the premise that instance-conditional value is the source of its gains. Lookahead does matter on the long, heterogeneous Qwen ladder: the one-step (myopic) variant drops to 0.654, while on the three-model cascades the two are within 0.003.

6 Why the Simulation Misled

Our earlier simulator drew correctness from an item-response model with RouterBench-style costs and hand-set categories whose value of escalation differed strongly. Three measured differences explain most of the gap. (1) **Baseline**: inside the simulator itself, replacing the global threshold by per-stage thresholds cut VoC-Stop's saving at the GPT-3.5 operating point from 71% to 25%. (2) **Category structure**: the simulator built in categories that reward conditioning; in real data category conditioning adds at most 0.013 AUCC to thresholds and hurts VoC-Stop in two families. (3) **Cost structure**: in the simulator the models before the largest cost 36% of it in total; in real Qwen1.5 they cost 82%, so a full-ladder cascade pays much more before escalating, which is why model selec-

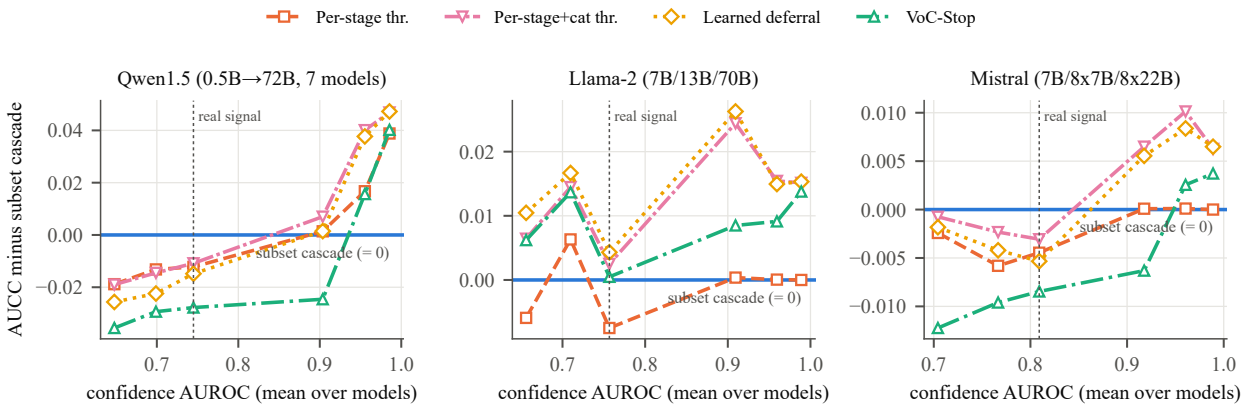


Figure 2: E2: AUCC of each rule minus the subset cascade as confidence quality varies. Points left of the dashed line are noised versions of the real signal; points to the right are label-sharpened (not deployable). On Qwen1.5 and Mistral, adaptive rules overtake simple model selection only around AUROC 0.9.

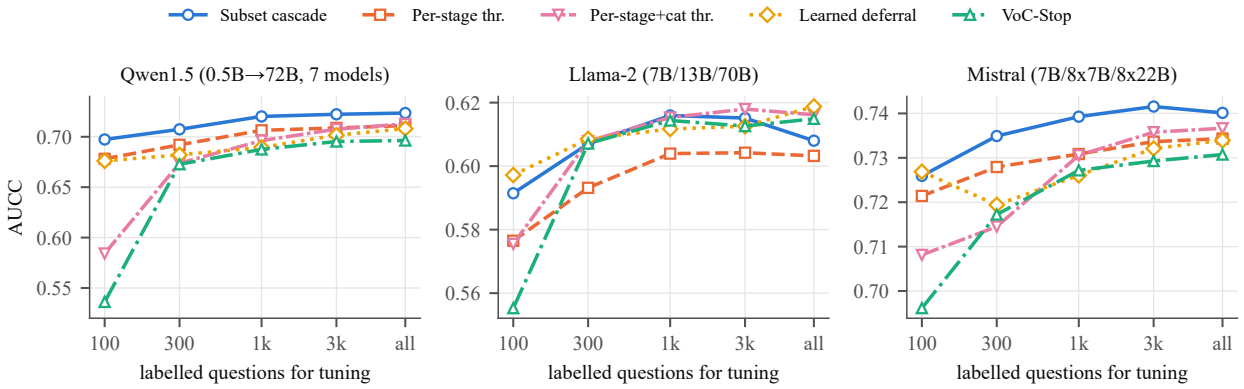


Figure 3: E3: AUCC vs. number of labelled questions used for tuning (3 splits). The subset cascade is near its full-data value with 100 labels; VoC-Stop and category thresholds need 300–3,000.

tion matters. Cross-model correctness correlation was *not* the culprit (0.14 in simulation vs. 0.19–0.45 in real data). The lesson generalises: evaluate deferral rules on real logs, and always include per-stage thresholds and single models as baselines.

7 Discussion and Guidance

For practitioners building a cascade with confidence signals like those studied here: (1) never tune a single global threshold; tune one per stage; (2) search over which models to include, since two well-chosen models often beat a full ladder; (3) compare against simply calling a single model, since on some families (here Llama-2 at high accuracy targets) no cascade saves cost; (4) spend effort on confidence quality, where the headroom is (Table 2), before adopting context-conditioned or lookahead rules, which are data-hungry and pay off mainly when confidence is strong.

8 Limitations

Models are 2023–24 open base models scored by option log-likelihood on multiple-choice tasks; modern instruction-tuned and API models, generative tasks and verbalised confidence may behave differently, and API models often expose no log-probabilities. Cost is modelled as active parameters, ignoring prompt length, hardware and provider pricing. Sharpened confidence in E2 is constructed from labels and only bounds what a better estimator could achieve. Subset search is limited to three models, we do not reimplement the full cascade-routing method [5], and results average 3–5 random splits.

9 Conclusion

On real logs from 13 models, the sophistication of the stopping rule matters far less than three simpler choices: tuning thresholds per stage, choosing which models to chain, and the quality of the confidence signal. We hope the released benchmark makes it easy to hold new deferral rules to this standard.

Table 2: E2: AUCC as confidence quality varies (3 splits). Sharpened signals use the label and are *not* deployable; they show what a better confidence estimator would buy. Best non-oracle in bold.

	Signal	AUROC	Per-stage	PS+cat	GBM	VoC-Stop	Subset
Qwen	noised (σ=2)	0.65	0.704	0.704	0.697	0.687	**0.723**
	noised (σ=1)	0.70	0.710	0.709	0.701	0.694	**0.723**
	real (max-prob)	0.74	0.712	0.713	0.709	0.696	**0.724**
	sharpened (β=0.5)	0.90	0.771	**0.777**	0.771	0.745	0.770
	sharpened (β=1)	0.96	0.821	**0.845**	0.842	0.820	0.804
	sharpened (β=2)	0.99	0.875	0.883	**0.884**	0.877	0.836
Llama-2	noised (σ=2)	0.66	0.592	0.604	**0.608**	0.604	0.597
	noised (σ=1)	0.71	0.602	0.610	**0.612**	0.609	0.596
	real (max-prob)	0.76	0.606	0.616	**0.618**	0.614	0.614
	sharpened (β=0.5)	0.91	0.671	0.695	**0.697**	0.679	0.671
	sharpened (β=1)	0.96	0.720	**0.736**	0.735	0.729	0.720
	sharpened (β=2)	0.99	0.740	0.755	**0.755**	0.753	0.740
Mistral	noised (σ=2)	0.70	0.723	0.725	0.724	0.713	**0.725**
	noised (σ=1)	0.77	0.728	0.731	0.729	0.724	**0.733**
	real (max-prob)	0.81	0.735	0.736	0.734	0.731	**0.740**
	sharpened (β=0.5)	0.92	0.776	**0.782**	0.781	0.769	0.776
	sharpened (β=1)	0.96	0.804	**0.814**	0.812	0.806	0.804
	sharpened (β=2)	0.99	0.822	0.829	**0.829**	0.826	0.822

Table 3: E4: task shift. Rules tuned on MMLU, evaluated on ARC-Challenge + WinoGrande (category information removed; 5 training subsamples).

	Qwen1.5 (7 models)		Llama-2 (3)		Mistral (3)	
Method	AUCC	Cost@99	AUCC	Cost@99	AUCC	Cost@99
Global threshold	0.634	–	0.693	97%	0.740	83%
Per-stage threshold	0.657	–	0.696	87%	0.745	73%
Per-stage + category thr.	0.657	–	0.696	87%	0.745	73%
VoC-Stop	0.659	–	**0.698**	91%	0.745	81%
Learned deferral (GBM)	0.671	–	0.697	93%	0.743	90%
Subset cascade ($\leq$3)	**0.695**	62%	0.673	100%	**0.746**	64%
Single models (no cascade)	0.695	100%	0.673	100%	0.738	100%
Oracle (upper bound)	0.835	3%	0.790	21%	0.820	22%

Reproducibility. All code, the data-fetching script and every result file are at `https://github.com/thejenilsoni/llm-cascade-audit`. A single script re-runs all experiments in about 40 minutes on a desktop PC, without API keys or GPUs.

Use of AI tools. AI-based coding and writing assistants were used during development of the code and preparation of the manuscript. The author designed the study, reviewed all code and results, and takes full responsibility for the content.

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Table 4: E5: ablation, adding one ingredient at a time (5 splits).

	Qwen1.5 (7 models)		Llama-2 (3)		Mistral (3)	
Method	AUCC	Cost@99	AUCC	Cost@99	AUCC	Cost@99
raw confidence, global thr.	0.662	164%	0.584	107%	0.726	98%
Global threshold	0.668	112%	0.584	111%	0.726	98%
Per-stage threshold	0.711	96%	0.603	109%	0.734	94%
VoC, one-step (myopic)	0.654	–	0.614	–	0.734	–
VoC, lookahead, no category	**0.713**	97%	0.612	108%	0.736	96%
VoC-Stop	0.696	–	0.615	–	0.731	122%
Per-stage + category thr.	0.713	101%	**0.616**	107%	**0.737**	93%

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